

Weather & Air Quality Station — Resources

Chain K — Hardware & Systems Foundations · further reading and videos, organized by lesson

Lesson 1 — Sensors & Real-World Data

READ [Random Nerd Tutorials — ESP32 with BME280 using Arduino IDE](#)

Random Nerd Tutorials — a widely-cited, established ESP32/Arduino tutorial site

Lesson 2 — Solar Power & Battery Sizing

WATCH [How To Size A Solar Charge Controller? \(MPPT & PWM\)](#)

KiloWatt Maker with Younes

Lesson 3 — Connectivity

READ [LoRaWAN vs Wi-Fi for IoT Sensors: Range, Power and Gateway Comparison](#)

Robustel

Lesson 4 — Weatherproofing & Reliability

READ [How to Design an IP65/IP67 Enclosure for Outdoor Electronics](#)

Ohmframe — same guide referenced by Walkie-Talkie-Build's waterproofing lesson